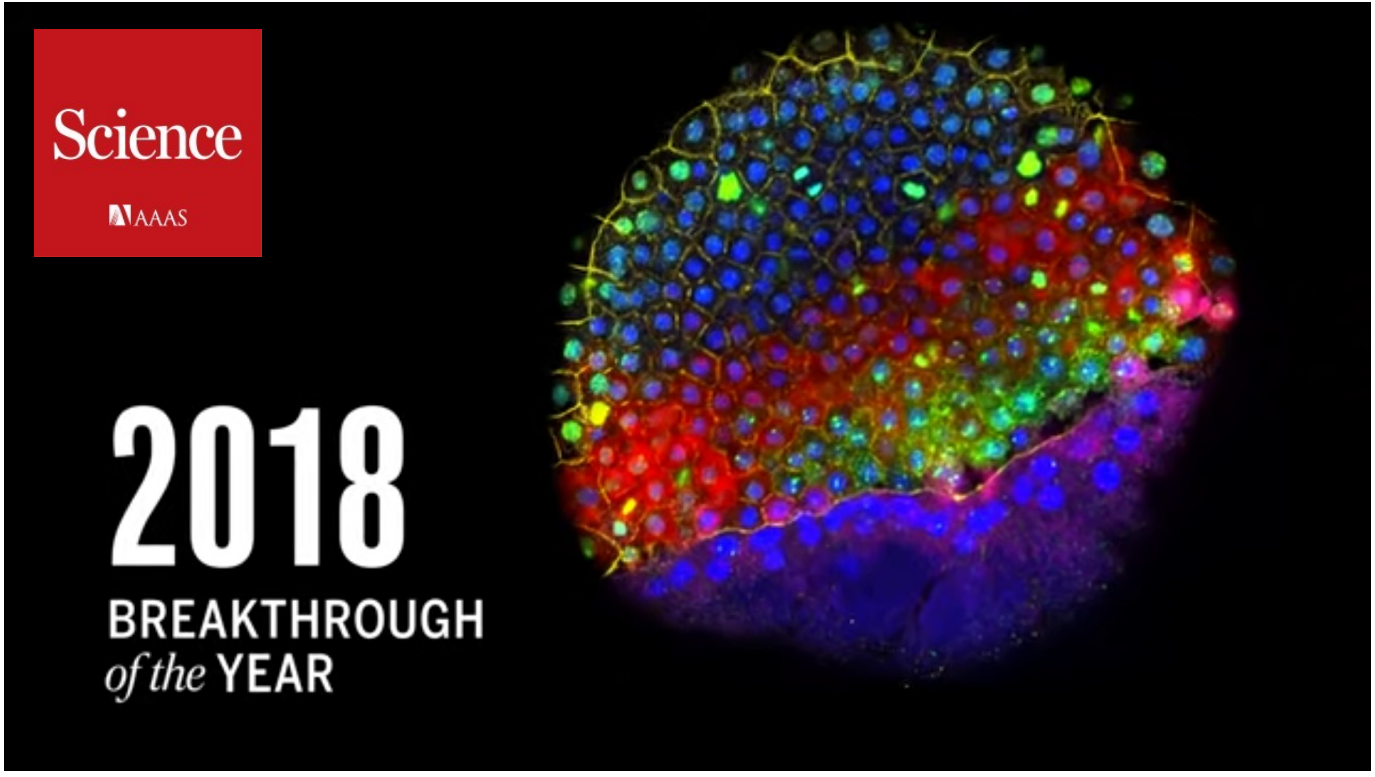
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Session 1: Foundations of Single-Cell RNA Sequencing**



@NikolayOskolkov

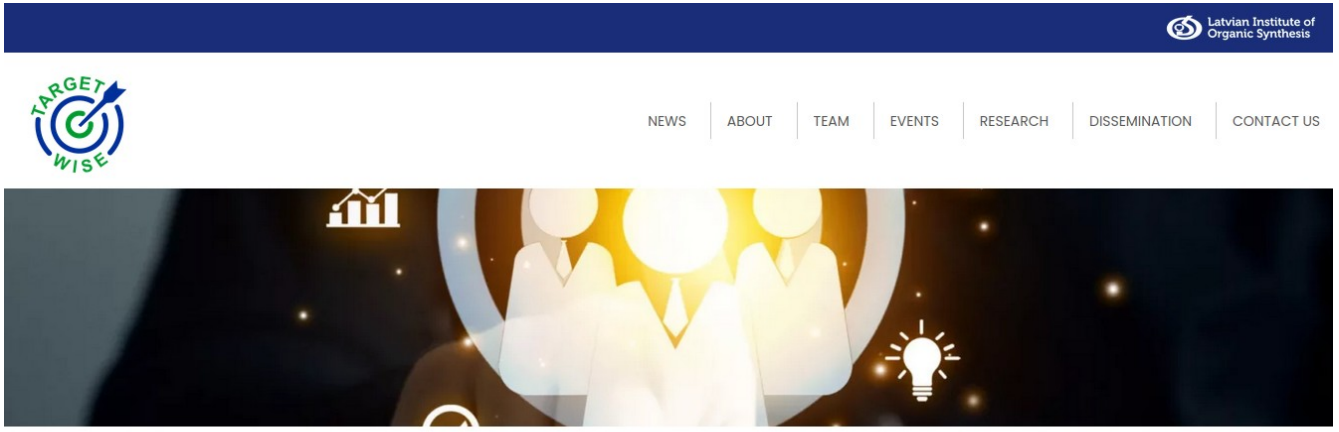
@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Metabolic Research Group at NIRI



1 more postdoctoral fellow and 1 PhD student to be hired

If you know anyone who might be interested, please contact me!



PhD. Nikolay Oskolkov
Group Leader (PI) of the Metabolic Research Group



Mohammad Banabazi, postdoc in Metabolic Research Group



Kristina Grausa, PhD student in Metabolic Research Group



Daniel Rivas, MD, postdoc in Metabolic Research Group

About us

Organizer: Oliver Hooker, PR Stats courses

oliver@prstats.org

Instructor:

Dr. Nikolay Oskolkov, MRG Group Leader, NIRI, Riga,
lecturer at Lund University, Lund,Sweden

nikolay.oskolkov@niri.lv



@NikolayOskolkov

@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Brief introduction: who am I

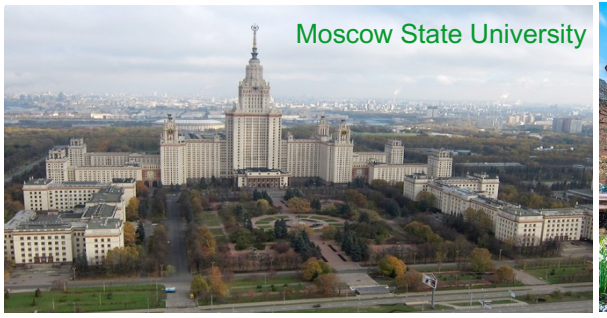
2007 PhD in theoretical physics in Moscow, Russia

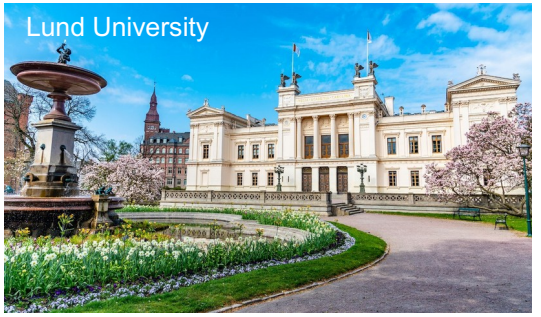
2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

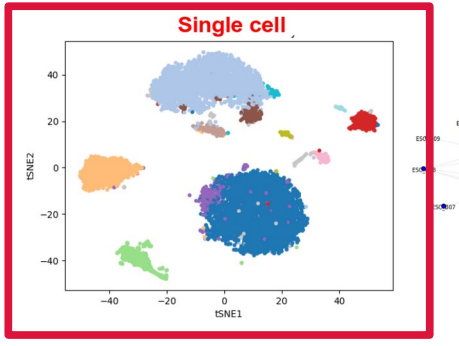
60+ publications; h-index = 28; 4,600+ citations

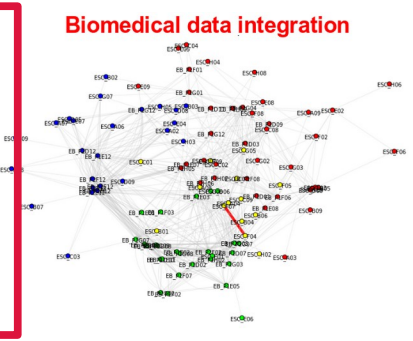


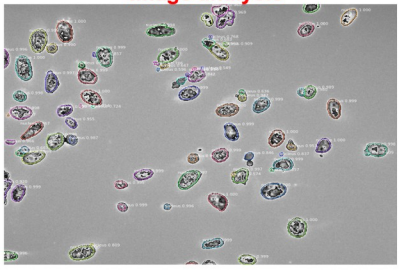


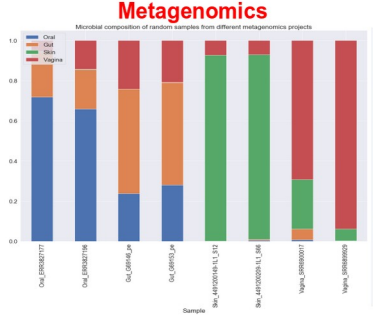


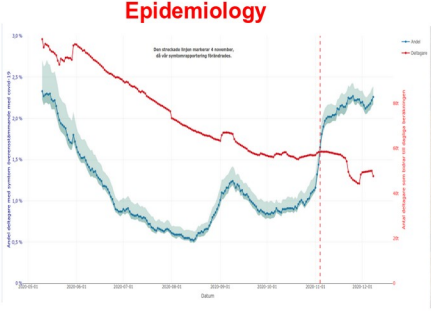














My experience with scRNAseq analysis

nature COMMUNICATIONS

ARTICLE

DOI: 10.1038/s41467-018-07582-3 OPEN

Spatially and functionally distinct subclasses of breast cancer-associated fibroblasts revealed by single cell RNA sequencing

Michael Bartoschek[1], Nikolay Oskolkov[2], Matteo Bocci[1], John Lövrot[3], Christer Larsson[1], Mikael Sommarin[4], Chris D. Madsen[1], David Lindgren[1], Gyula Pekar[5], Göran Karlsson[4], Markus Ringnér[2], Jonas Bergh[3], Åsa Björklund[6] & Kristian Pietras[1]

Cancer-associated fibroblasts (CAFs) are a major constituent of the tumor microenvironment, although their origin and roles in shaping disease initiation, progression and treatment response remain unclear due to significant heterogeneity. Here, following a negative selection strategy combined with single-cell RNA sequencing of 768 transcriptomes of mesenchymal cells from a genetically engineered mouse model of breast cancer, we define three distinct subpopulations of CAFs. Validation at the transcriptional and protein level in several experimental models of cancer and human tumors reveal spatial separation of the CAF subclasses attributable to different origins, including the peri-vascular niche, the mammary fat pad and the transformed epithelium. Gene profiles for each CAF subtype correlate to distinctive functional programs and hold independent prognostic capability in clinical cohorts by association to metastatic disease. In conclusion, the improved resolution of the widely defined CAF population opens the possibility for biomarker-driven development of drugs for precision targeting of CAFs.

Oskolkov *et al. Skeletal Muscle* (2022) 12:16
https://doi.org/10.1186/s13395-022-00299-4

SciLifeLab
Skeletal Muscle

RESEARCH Open Access

High-throughput muscle fiber typing from RNA sequencing data

Nikolay Oskolkov[1,2], Malgorzata Santel[3], Hemang M. Parikh[4], Ola Ekström[1], Gray J. Camp[3], Eri Miyamoto-Mikami[5], Kristoffer Ström[1,6], Bilal Ahmad Mir[1], Dmytro Kryvokhyzha[1], Mikko Lehtovirta[1,7], Hiroyuki Kobayashi[8], Ryo Kakigi[9], Hisashi Naito[5], Karl-Fredrik Eriksson[1], Björn Nystedt[10], Noriyuki Fuku[5], Barbara Treutlein[3], Svante Pääbo[3,11] and Ola Hansson[1,7*]

Abstract

Background: Skeletal muscle fiber type distribution has implications for human health, muscle function, and performance. This knowledge has been gathered using labor-intensive and costly methodology that limited these studies. Here, we present a method based on muscle tissue RNA sequencing data (totRNAseq) to estimate the distribution of skeletal muscle fiber types from frozen human samples, allowing for a larger number of individuals to be tested.

Methods: By using single-nuclei RNA sequencing (snRNAseq) data as a reference, cluster expression signatures were produced by averaging gene expression of cluster gene markers and then applying these to totRNAseq data and inferring muscle fiber nuclei type via linear matrix decomposition. This estimate was then compared with fiber type distribution measured by ATPase staining or myosin heavy chain protein isoform distribution of 62 muscle samples in two independent cohorts ($n = 39$ and 22).

Results: The correlation between the sequencing-based method and the other two were $r_{ATPas} = 0.44$ [0.13–0.67], [95% CI], and $r_{myosin} = 0.83$ [0.61–0.93], with $p = 5.70 \times 10^{-3}$ and 2.00×10^{-6}, respectively. The deconvolution inference of fiber type composition was accurate even for very low totRNAseq sequencing depths, i.e., down to an average of ~ 10,000 paired-end reads.

Conclusions: This new method (https://github.com/OlaHanssonLab/PredictFiberType) consequently allows for measurement of fiber type distribution of a larger number of samples using totRNAseq in a cost and labor-efficient way. It is now feasible to study the association between fiber type distribution and e.g. health outcomes in large well-powered studies.

Introduction

Our bodies constitute to ~ 30–40% of the skeletal muscle, and it is the most abundant form of the three types of muscle, the others being smooth and cardiac. The skeletal muscle is composed of different fiber types (i.e., muscle cell types), and the relative proportions of these types vary among the muscles, locations within the muscles, individuals, and the sex of individuals [1–4]. The oxidative and glycolytic potential and the contractile properties differ considerably between fiber types, with the mitochondria-rich slow-twitch fibers (type I) having higher oxidative capacity, and fast-twitch fibers (type IIa and type IIx) having higher glycolytic capacity [1]. The proportions also change as people age, with type II fibers being preferentially affected by sarcopenia [5]. Exercising the skeletal muscle is a major site for catabolic metabolism of the blood glucose and lipids and the metabolic characteristics of this tissue influence both the

*Correspondence: Ola.Hansson@med.lu.se
[1] Department of Clinical Sciences, Lund University, Malmö, Sweden
Full list of author information is available at the end of the article

BMC

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scRNAseq Cancer-Associated Fibroblasts

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Spatially and functionally distinct subclasses of breast cancer-associated fibroblasts revealed by single cell RNA sequencing

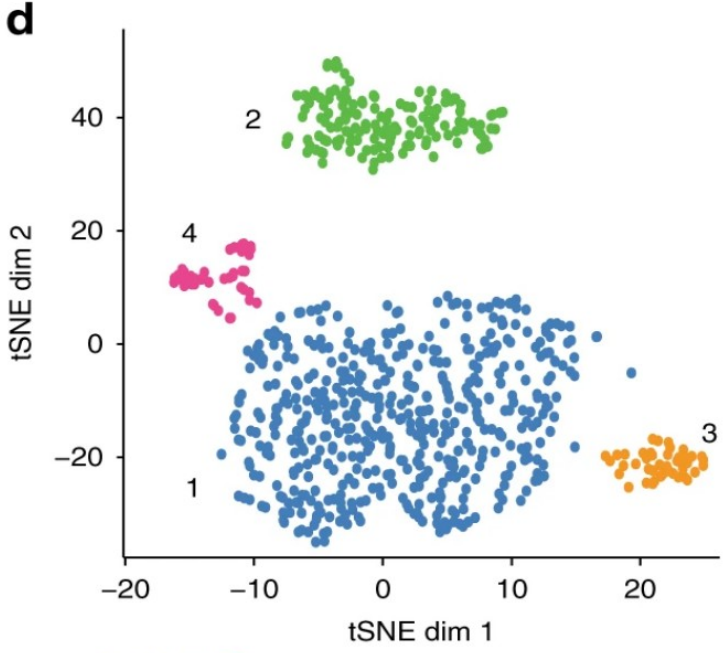
Michael Bartoschek, Nikolay Oskolkov, Matteo Bocci, John Lövrot, Christer Larsson, Mikael Sommar, Chris D. Madsen, David Lindgren, Gyula Pekar, Göran Karlsson, Markus Ringnér, Jonas Bergh, Åsa Björklund & Kristian Pietras

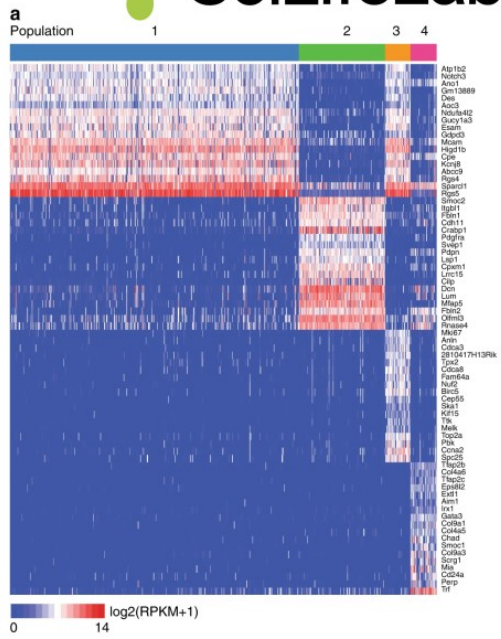
Nature Communications **9**, Article number: 5150 (2018) | Cite this article

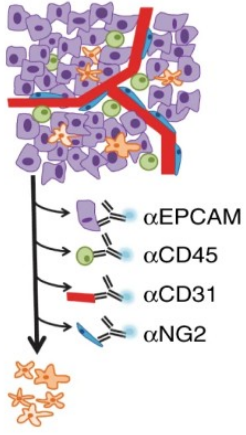
49k Accesses | **495** Citations | **91** Altmetric | Metrics

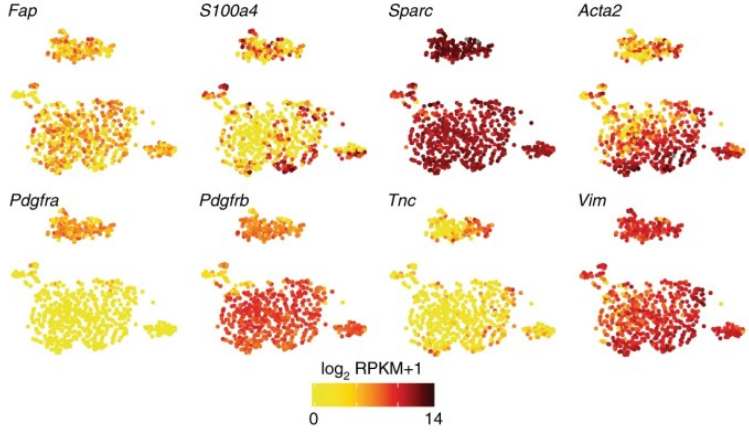
Abstract

Cancer-associated fibroblasts (CAFs) are a major constituent of the tumor microenvironment, although their origin and roles in shaping disease initiation, progression and treatment response remain unclear due to significant heterogeneity. Here, following a negative selection strategy combined with single-cell RNA sequencing of 768 transcriptomes of mesenchymal cells from a genetically engineered mouse model of breast cancer, we define three distinct subpopulations of CAFs. Validation at the transcriptional and protein level in several experimental models of cancer and human tumors reveal spatial separation of the CAF subclasses attributable to different origins, including the peri-vascular niche, the mammary fat pad and the transformed epithelium. Gene profiles for each CAF subtype correlate to distinctive functional programs and hold independent prognostic capability in









Evolutionary single cell: Human vs. Chimp



Gray Camp and Barbara Treutlein,
University of Basel

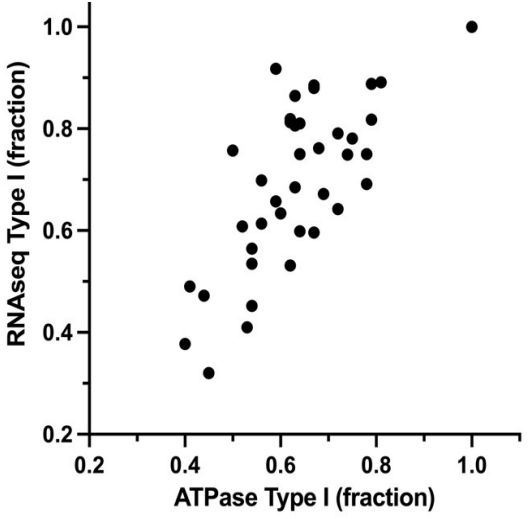


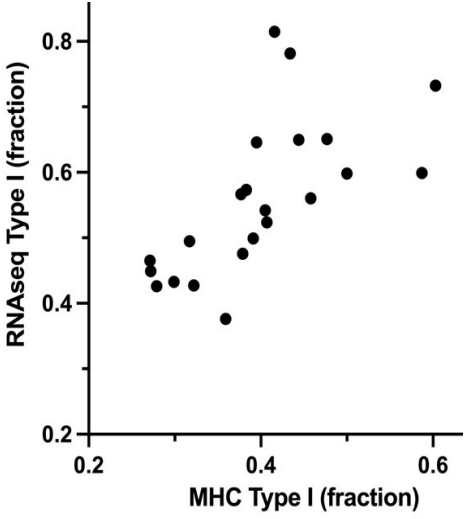
Svante Pääbo,
Max Plank Leipzig

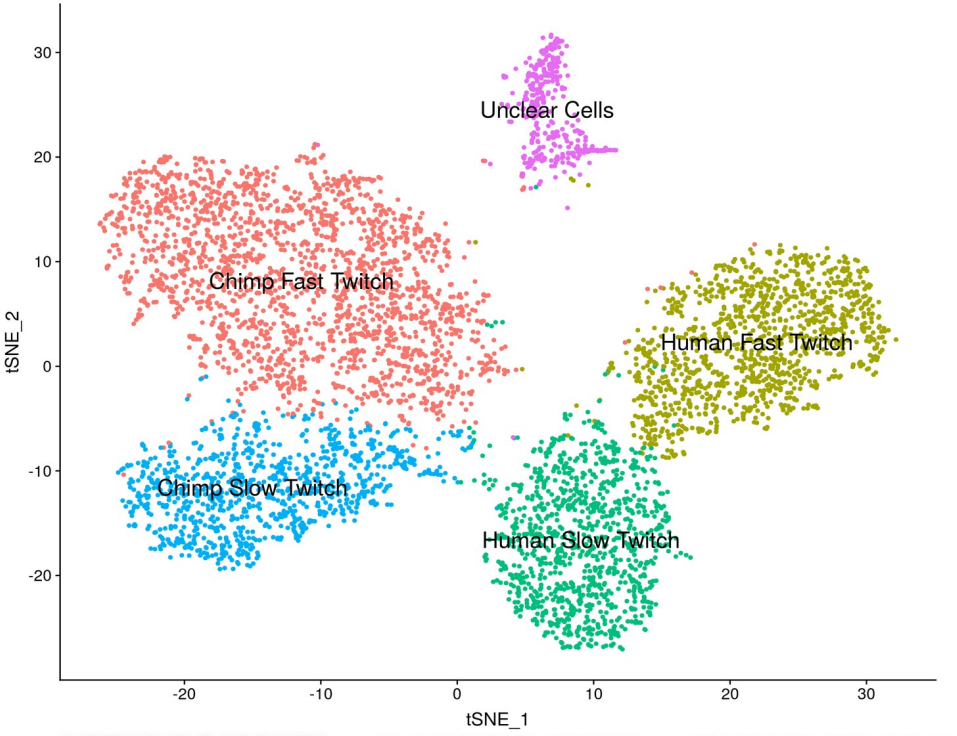


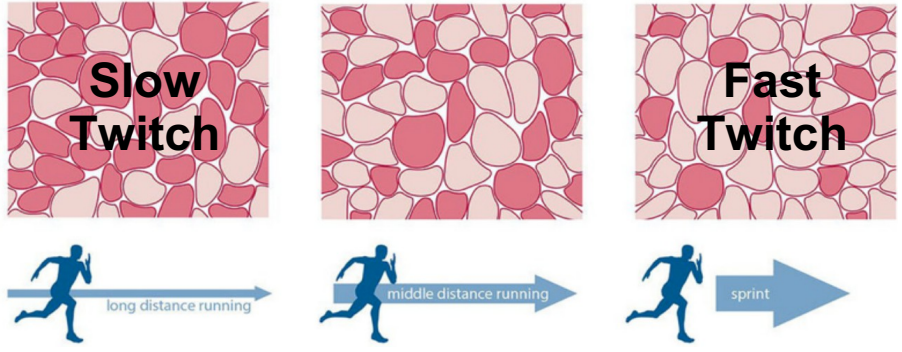
Ola Hansson,
Lund University

De-convolution prediction of fiber-type composition









Session content

Topics we'll cover in this session:

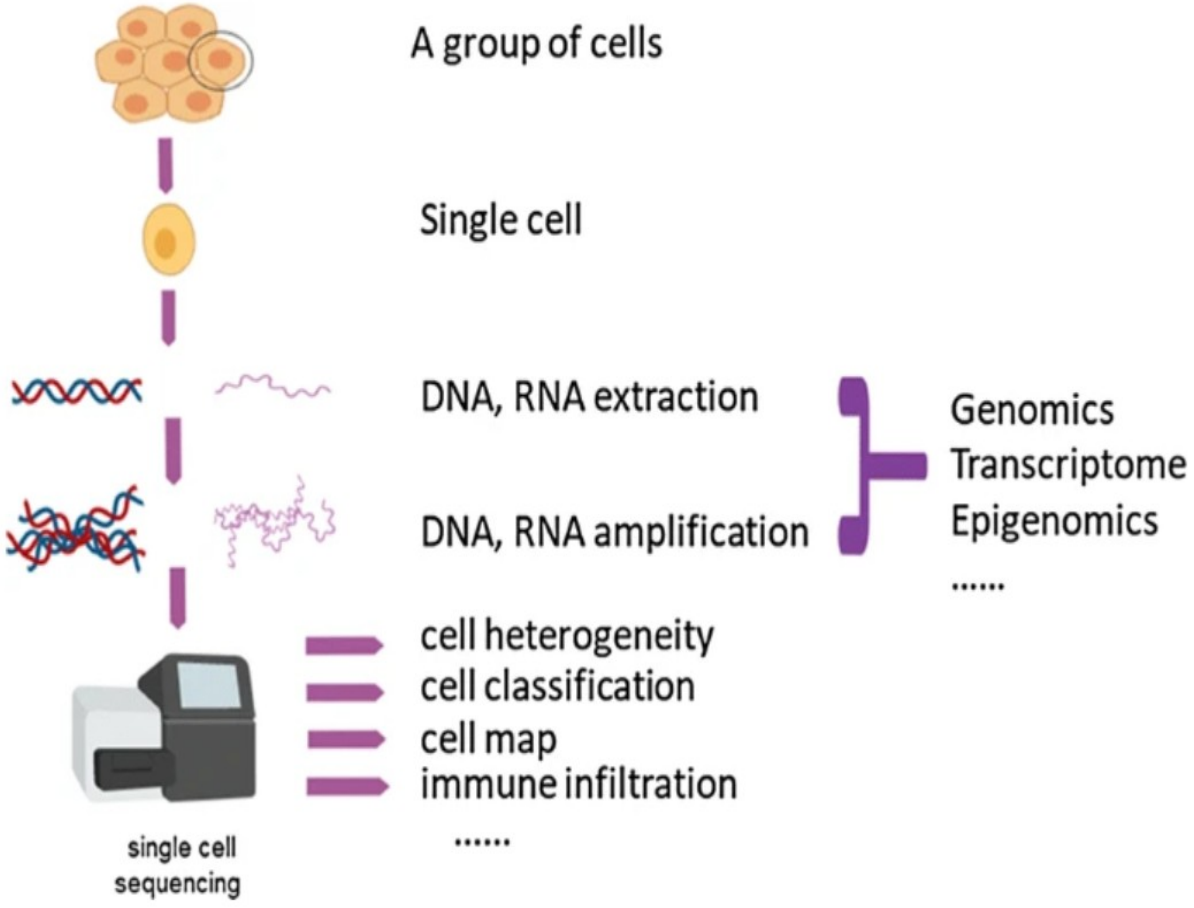
1) Recent developments in single cell omics technologies

2) Big data in single cell omics

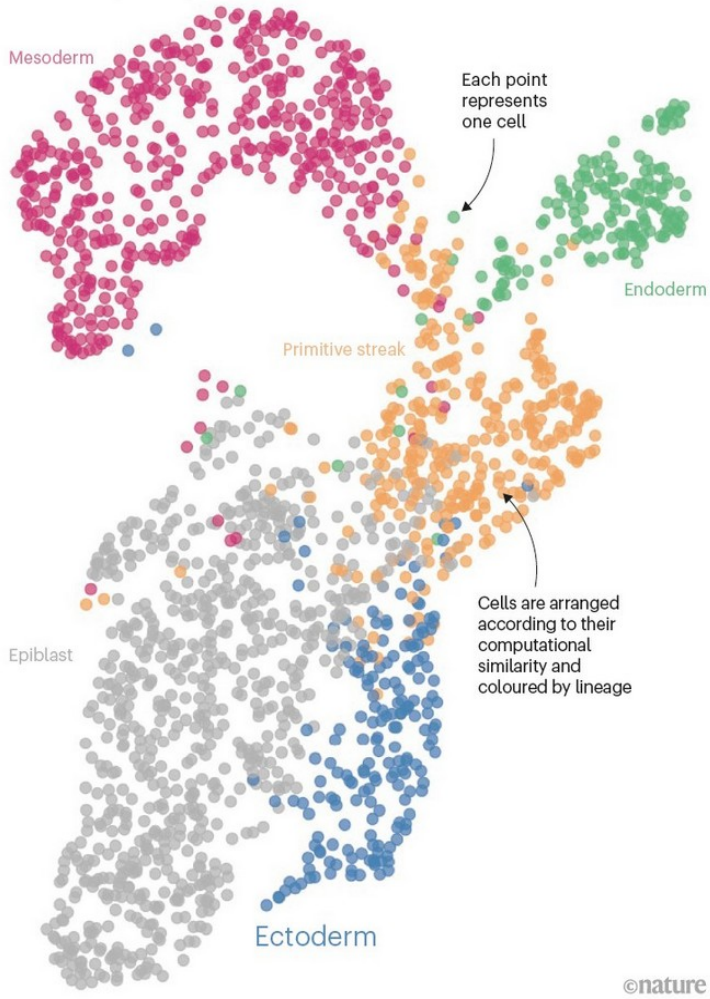
3) The curse of dimensionality and promise of AI

4) Overview of main steps of scRNAseq computational analysis

Recent Advances in Single Cell Biology



The principle of single-cell sequencing. It is a process of isolating a single cell for sequencing and studying cell heterogeneity, molecular mapping, immune infiltration and epigenetic changes



scRNAseq workflow: cell isolation / capture

I. Tissue Procurement

Source:
- Primary human
- Model organism
- Cell culture

Key considerations:
- Biological variation
- Sampling/handling variation
- Duration of sourcing

Study design:
- Biological replicates
- Technical replicates
- Cell number calculation
- Workflow optimization

II. Tissue Dissociation

Method:
- Mechanical mincing
- Enzymatic digestion
- Automated blending
- Microfluidics devices

Key considerations:
- Experimental consistency
- Shortest duration
- Highest cell/nucleus quality
- Representation of all cell types

Quality control:
- FACS analysis
- qPCR for marker genes
- Imaging of cell integrity
- RNA quality (RIN)

III. Cell Enrichment (optional)

Method:
- Differential centrifugation, sedimentation, filtration
- Antibody labeling for positive/negative selection
- Flow cytometry or bead-based enrichment
- Dead cell removal

Key considerations:
- Additional handling
- Longer duration
- Loss of RNA quality
- Transcriptome changes

IV. Single Cell RNAseq Platform

Method:
- Droplet-based
- Tube-based after FACS
- Microwell-based
- Microfluidics-enabled

Key considerations:
- Cell throughput and handling time
- Gene coverage and cell type detection
- Whole transcript versus 3'end counting
- Imaging capability for doublet detection

V. Library Sequencing

Method:
- Illumina NGS
- Compatible with cDNA library

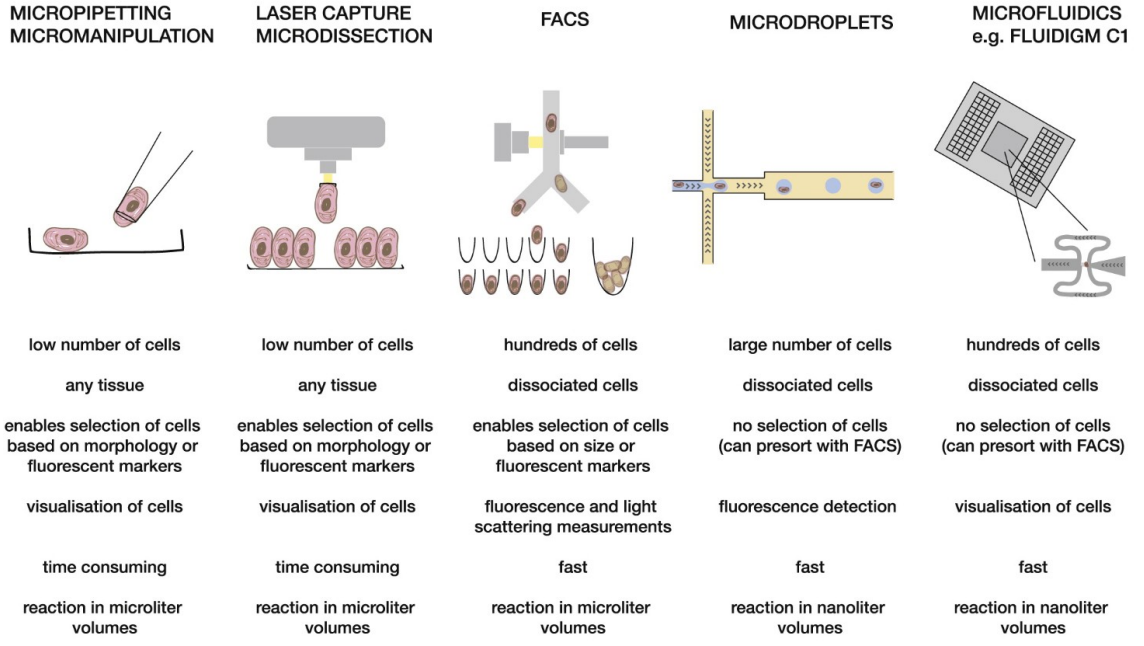
Sequencing depth considerations:
- 3'end counting: low depth ~50K RPC
- Whole transcript: high depth ~1M RPC
- Alternative splicing: ~20-30M RPC
- Iterative optimization for biological system

VI. Computational Analysis

Key considerations:
- Separation of *batch* and *condition*
- Technical vs. biological variation

Sample Batch correction approaches:
- Cell Hashing
- Demuxlet
- Canonical correlation analysis (CCA)
- MAST

Nguyen et al., Frontiers in Cell and Developmental Biology 2018



Kolodziejczyk A et al, Molecular Cell, 2015

- ✓ Isolate cells in compartments
- ✓ Add UMI & barcode for later pooling
- ✓ Amplify
- ✓ Fragment (e.g Tn5 tagmentation)
- ✓ Sequence pooled libraries
- ✓ Analyze your data!

Short history of scRNAseq methods

PERSPECTIVE

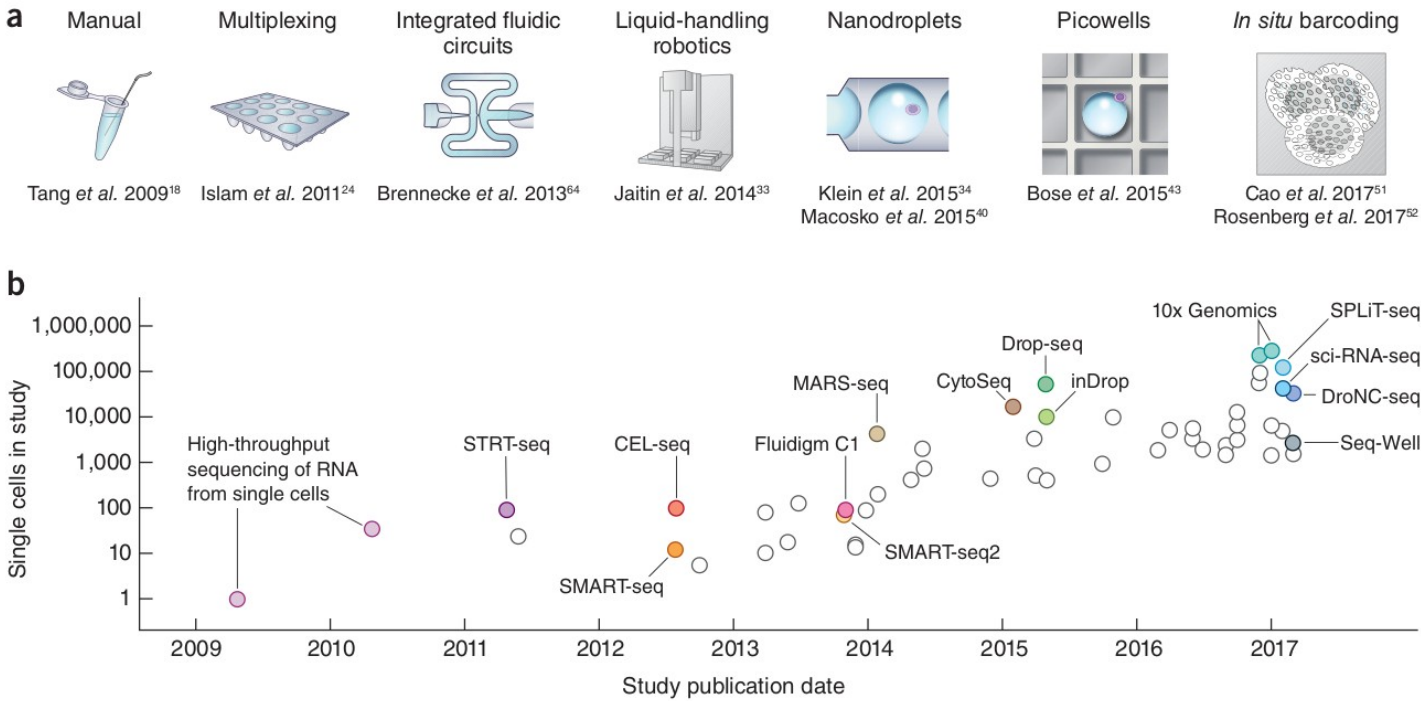


Figure 1 | Scaling of scRNA-seq experiments. (**a**) Key technologies that have allowed jumps in experimental scale. A jump to ~100 cells was enabled by sample multiplexing, and then a jump to ~1,000 cells was achieved by large-scale studies using integrated fluidic circuits, followed by a jump to several thousands of cells with liquid-handling robotics. Further orders-of-magnitude increases bringing the number of cells assayed into the tens of thousands were enabled by random capture technologies using nanodroplets and picowell technologies. Recent studies have used *in situ* barcoding to inexpensively reach the next order of magnitude of hundreds of thousands of cells. (**b**) Cell numbers reported in representative publications by publication date. Key technologies are indicated. A full table with corresponding numbers is available as **Supplementary Table 1**.

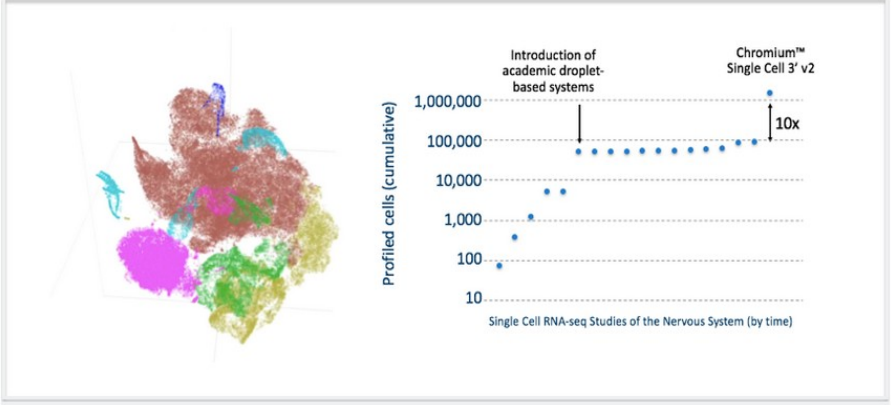
V. Svensson et al., Nature Protocols 2018

Big Data in single cell genomics

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« Back to Blog ‹ Newer Article Older Article ›



Our 1.3 million single cell dataset is ready to download

0 KUDOS

POSTED BY: grace-10x, on Feb 21, 2017 at 2:28 PM

At ASHG last year, we announced our 1.3 Million Brain Cell Dataset, which is, to date, the largest dataset published in the single cell RNA-sequencing (scRNA-seq) field. Using the Chromium™ Single Cell 3' Solution (v2 Chemistry), we were able to sequence and profile 1,308,421 individual cells from embryonic mice brains. Read more in our application note Transcriptional Profiling of 1.3 Million Brain Cells with the Chromium™ Single Cell 3' Solution.

Available online at www.sciencedirect.com

ScienceDirect

Current Opinion in Systems Biology

Single cells make big data: New challenges and opportunities in transcriptomics

Philipp Angerer[1], Lukas Simon[1], Sophie Tritschler[1], F. Alexander Wolf[1], David Fischer[1] and Fabian J. Theis[1,2]

Abstract

Recent technological advances have enabled unprecedented insight into transcriptomics at the level of single cells. Single cell transcriptomics enables the measurement of transcriptomic information of thousands of single cells in a single experiment. The volume and complexity of resulting data make it a paradigm of big data. Consequently, the field is presented with new scientific and, in particular, analytical challenges where currently no scalable solutions exist. At the same time, exciting opportunities arise from increased resolution of single-cell RNA sequencing data and improved statistical power of ever growing datasets. Big single cell RNA sequencing data promises valuable insights into cellular heterogeneity which may significantly improve our understanding of biology and human disease. This review focuses on single cell transcriptomics and highlights the inherent opportunities and challenges in the context of big data analytics.

Addresses

[1] Institute of Computational Biology, Helmholtz Center Munich, German Research Center for Environmental Health, Neuherberg, Germany
[2] Department of Mathematics, Technical University of Munich, Garching, Germany

Corresponding author: Theis, Fabian J. Institute of Computational Biology, Helmholtz Center Munich, German Research Center for Environmental Health, Neuherberg, Germany. (fabian.theis@helmholtz-muenchen.de)

Current Opinion in Systems Biology 2017, 4:85–91

This review comes from a themed issue on **Big data acquisition and**

volume — the amount of data, velocity — the required processing speed, veracity — trustworthiness and availability, and variety — necessary model complexity [2]. The traditional scientific big data field is astronomy because of the huge *volume* of image data produced by telescopes with a high daily *velocity* [3]. Big data has also reached biology, mainly driven through the advent of next generation sequencing technology. For biologists, assessing *veracity* through statistical means is nothing new.

Recent technological advances now allow the profiling of single cells at a *variety* of omic layers (genomes, epigenomes, transcriptomes and proteomes) at an unprecedented level of resolution [4]. Single cell transcriptomics (SCT) entails the profiling of all messenger RNAs present in a single cell and constitutes the most widely-used sc profiling technology [4]. Unlike bulk RNA-seq profiling where sequencing libraries are generated from thousands of cells, scRNA-seq technologies isolate single cells and generate cell-specific sequencing libraries (e.g. Fluidigm [5]) mark RNA content with a cell-specific molecular barcode [6–9]. Both approaches generate gene expression estimates at the single cell level [10]. SCTenables, for the first time, the measurement of the transcriptomic information of thousands, and up to millions of single cells, in a single experiment [7]. The complexity of SCT data coupled with the massive volume inherent to next generation sequencing data makes it a paradigm of big data.

AI applications in single cell biology

nature methods
ARTICLES
https://doi.org/10.1038/s41592-018-0229-2

Deep generative modeling for single-cell transcriptomics

Romain Lopez[1], Jeffrey Regier[1], Michael B. Cole[2], Michael I. Jordan[1,3] and Nir Yosef[1,4,5*]

Single-cell transcriptome measurements can reveal unexplored biological diversity, but they suffer from technical noise and bias that must be modeled to account for the resulting uncertainty in downstream analyses. Here we introduce single-cell variational inference (scVI), a ready-to-use scalable framework for the probabilistic representation and analysis of gene expression in single cells (https://github.com/YosefLab/scVI). scVI uses stochastic optimization and deep neural networks to aggregate information across similar cells and genes and to approximate the distributions that underlie observed expression values, while accounting for batch effects and limited sensitivity. We used scVI for a range of fundamental analysis tasks including batch correction, visualization, clustering, and differential expression, and achieved high accuracy for each task.

nature methods

Article
https://doi.org/10.1038/s41592-024-02201-0

scGPT: toward building a foundation model for single-cell multi-omics using generative AI

Received: 12 July 2023
Accepted: 30 January 2024
Published online: 26 February 2024
Check for updates

Haotian Cui[1,2,3,8], Chloe Wang[1,2,3,8], Hassaan Maan[1,3,4], Kuan Pang[2,3], Fengning Luo[2,3], Nan Duan[5] & Bo Wang[1,2,3,4,6,7]

Generative pretrained models have achieved remarkable success in various domains such as language and computer vision. Specifically, the

nature COMMUNICATIONS

ARTICLE
DOI: 10.1038/s41467-018-04368-5 OPEN

Interpretable dimensionality reduction of single cell transcriptome data with deep generative models

Jiarui Ding[1,2,3,4], Anne Condon[1] & Sohrab P. Shah[1,2,3,5]

Huang et al. Genome Biology (2023) 24:259
https://doi.org/10.1186/s13059-023-03100-x

Genome Biology

RESEARCH Open Access

Evaluation of deep learning-based feature selection for single-cell RNA sequencing data analysis

Hao Huang[1,2,3], Chunlei Liu[1,3], Manoj M. Wagle[1,2,3] and Pengyi Yang[1,2,3,4*]

Biomedicine & Pharmacotherapy 165 (2023) 115077

Contents lists available at ScienceDirect

Biomedicine & Pharmacotherapy

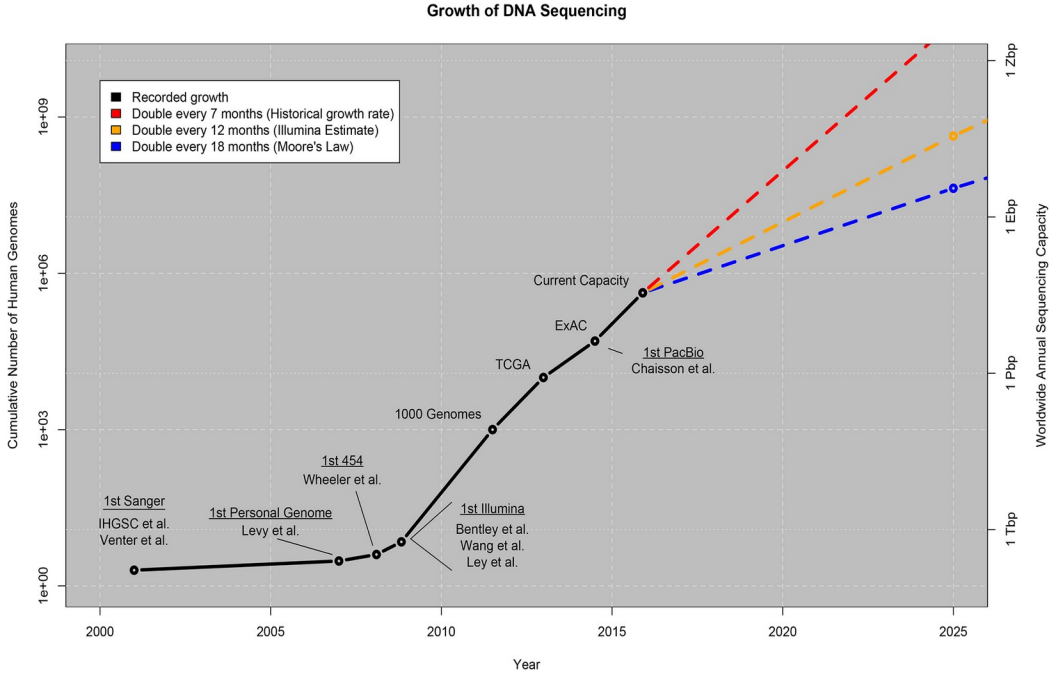
journal homepage: www.elsevier.com/locate/biopha

Review

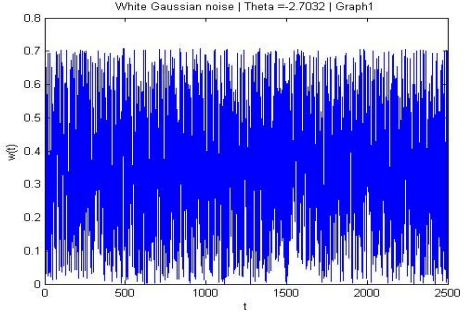
Deep learning applications in single-cell genomics and transcriptomics data analysis

Nafiseh Erfanian[a], A. Ali Heydari[b,c], Adib Miraki Feriz[a], Pablo Iañez[d], Afshin Derakhshani[e], Mohammad Ghasemigol[f], Mohsen Farahpour[g], Seyyed Mohammad Razavi[g], Saeed Nasseri[h], Hossein Safarpour[h,*], Amirhossein Sahebkar[i,j,k,**]

scRNAseq: Big Data vs. Little Data



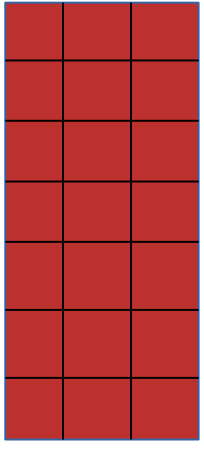
Stephens et al., (2015). Big Data: Astronomical or Genomical? PLoS Biology 13(7)



A file with **White Noise** can also take a lot of disk space

Genomics / WGS: **Little Data**

$N_1 \sim 10^3$

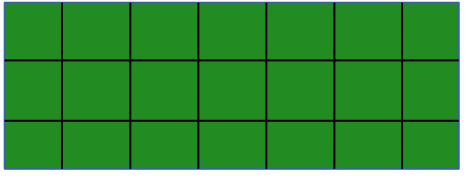


$P_1 \sim 10^6$

$N_1 * P_1 = N_2 * P_2 = 10^9$

scRNAseq: **Big Data**

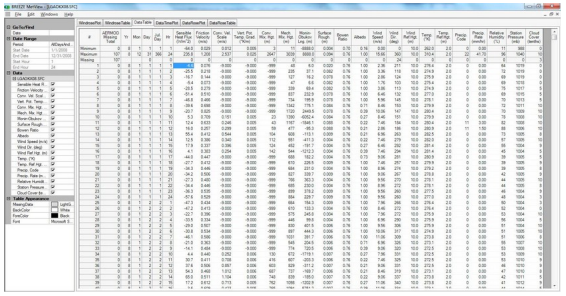
$N_2 \sim 10^6$



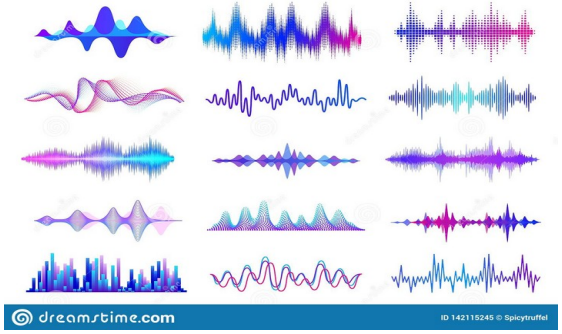
$P_2 \sim 10^3$

Various types of data around us

Tabular



Sound



Image



Text



Editing Wikipedia articles on

Medicine

Editing Wikipedia can be daunting for newbies, especially if you're contributing to Wikipedia for the first time as a class assignment. This guide is designed to assist students who have been assigned to contribute biomedical-related content to Wikipedia. Here's what other editors will expect you to know.

Be accurate

You're editing a resource millions of people use to make medical decisions, so it's vitally important to be accurate. Wikipedia is used more for medical information than the websites for WebMD, NIH, and the WHO. But with great power comes great responsibility!

Understand the guidelines

Wikipedia editors in the medicine area have developed additional guidelines to ensure that the content on Wikipedia is medically sound. Take extra time to read and understand these guidelines. When you edit an article, ensure your changes meet these special requirements. If not, your work is likely to be undone by other editors as they clean up after you. That takes valuable volunteer time away from creating content. If you're not comfortable working under these guidelines, talk to your instructor about an alternative off-wiki assignment.

Engage with editors

Part of the Wikipedia experience is receiving and responding to feedback from other editors. Do not submit your content on the last day, then leave Wikipedia! Real human volunteers from the Wikipedia community will likely read and respond to it, and it would be polite for you to acknowledge the time they volunteer to polish your work! Everything submitted to Wikipedia is reviewed by multiple, real humans! You may not get a comment, but if you do, please acknowledge it.

Watch out for close paraphrasing

Plagiarizing or close paraphrasing is never okay on Wikipedia and is a violation of your university's academic honor code. It's even worse on Wikipedia, as valuable volunteer time that could be used to create good content is instead used to clean up plagiarized work.

If you plagiarize or too closely paraphrase on Wikipedia, it is extremely likely that you'll be caught by other editors and there will be an online record of your plagiarism tied to your permanent online record.

Note that even educational materials from organizations like the WHO and abstracts of articles in PubMed are under copyright and cannot be copied. Write them in your own words whenever possible. If you aren't clear on what close paraphrasing is, visit your university's writing center.

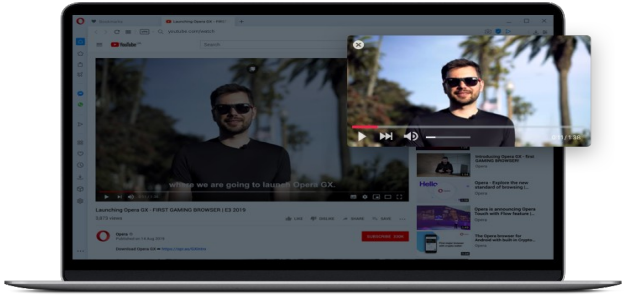
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Everybody on Wikipedia wants to make the best encyclopedia they can. Take the time to understand the rules, and soon you'll be contributing to a valuable resource you use on a daily basis!

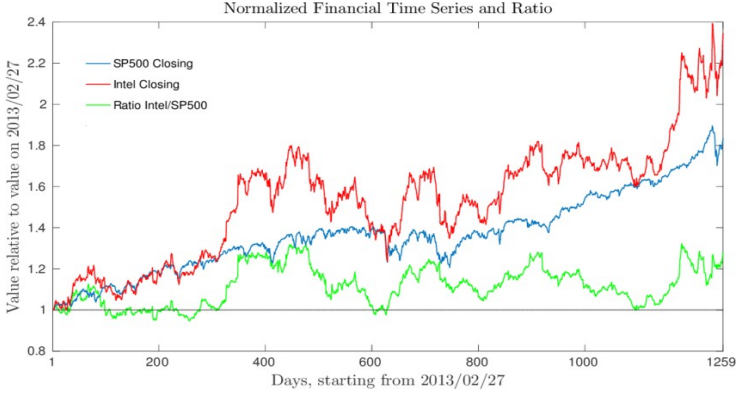
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DATA

Video



Time Series



Biological Data are High Dimensional

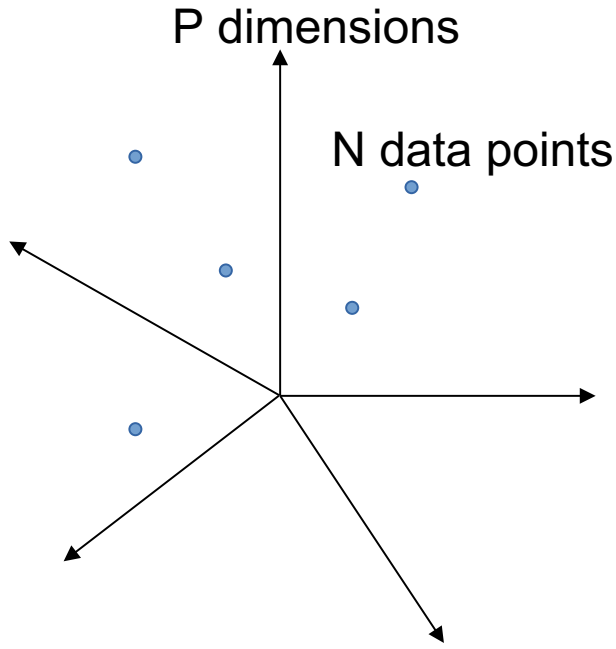
Statistical observations:
e.g. samples, cells etc.

N

P

0	3	1	0	2	3	8	1	1	3
1	1	0	0	7	1	2	2	3	3
1	2	2	0	0	6	7	1	2	2
1	2	3	10	0	4	6	1	0	5
3	2	2	1	4	3	2	1	6	0
7	4	4	5	3	9	6	1	6	1
7	1	1	5	2	8	9	1	3	6
5	0	1	6	2	0	0	0	1	5
1	6	3	3	4	6	2	0	1	1
1	2	2	4	1	1	3	0	8	2

Features: genes, proteins, microbes, metabolites etc.



High Dimensional Data:
P >> N

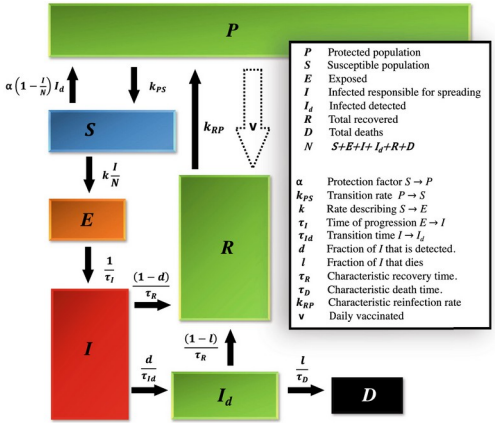
For a robust statistical analysis, one should properly “sample” the P-dimensional space, hence large sample size is required, N >> P

Some types of data analysis in Life Sciences

P is the number of features (genes, proteins, genetic variants etc.)
N is the number of observations (samples, cells, nucleotides etc.)

Biology / Biomedicine

Mathematical modeling



N ⟶ 0

Bayesianism



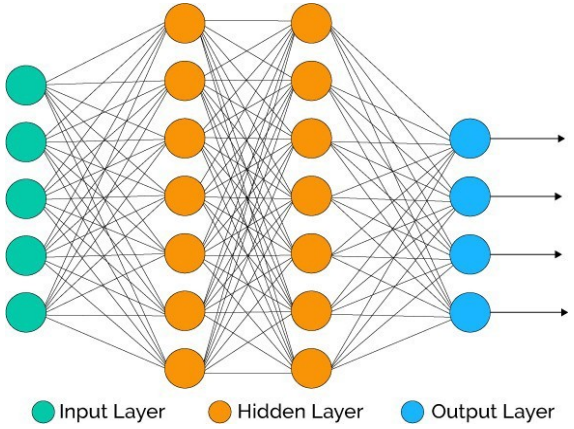
N << P

Frequentism



N ≈ P

Machine Learning



N >> P

Hypothesis-driven ⟵ **Amount of Data** ⟶ **Data-driven**

The Curse of Dimensionality

Ex.1

$$Y = \alpha + \beta X$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$(X^T X)^{-1} \sim \frac{1}{\det(X^T X)} \cdots \to \infty, \quad n << p$$

Ex.2 $$E[\hat{\sigma}^2] = \frac{n-p}{n}\sigma^2$$

Biased ML variance estimator in HD-space

Literature on the Curse of Dimensionality

POINTS OF SIGNIFICANCE

The curse(s) of dimensionality

There is such a thing as too much of a good thing.

Naomi Altman and Martin Krzywinski

We generally think that more information is better than less. However, in the 'big data' era, the sheer number of variables that can be collected from a single sample can be problematic. This embarrassment of riches is called the 'curse of dimensionality'[1] (CoD) and manifests itself in a variety of ways. This month, we discuss four important problems of dimensionality as it applies to data sparsity[1,2], multicollinearity[3], multiple testing[4] and overfitting[5]. These effects are amplified by poor data quality, which may increase with the number of variables.

Throughout, we use n to indicate the sample size from the population of interest and p to indicate the number of observed variables, some of which may have missing values for some samples. For example, we may have $n = 1{,}000$ subjects and $p = 200{,}000$ single-nucleotide polymorphisms (SNPs).

First, as the dimensionality p increases, the 'volume' that the samples may occupy grows rapidly. We can think of each of the n

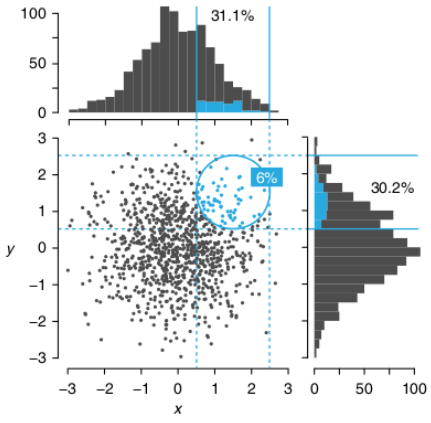


Fig. 1 | Data tend to be sparse in higher dimensions. Among 1,000 (x, y) points in which both x and y are normally distributed with a mean of 0 and s.d. $\sigma = 1$, only 6% fall within σ of $(x, y) = (1.5, 1.5)$ (blue circle). However, when the data are projected into a lower dimension—shown by histograms—about 30% of the points (all bins within blue solid lines) are within σ of 1.5. Blue

A and 100 to have the minor allele a. If we tabulate on two SNPs, A and B, we will expect only ten samples to exhibit both minor alleles with genotype ab. With SNPs A, B and C, we expect only one sample to have genotype abc, and with four or more SNPs, we expect empty cells in our table. We need a much larger sample size to observe samples with all the possible genotypes. As p increases, we may quickly find that there are no samples with similar values of a predictor.

Even with just five SNPs, our ability to predict and classify the samples is impeded because of the small number of subjects that have similar genotypes. In situations where there are many gene variants, this effect is exacerbated, and it may be very difficult to find affected subjects with similar genotypes and hence to predict or classify on the basis of genetic similarity.

If we treat the distance between points (e.g., Euclidian distance) as a measure of similarity, then we interpret greater distance as greater dissimilarity. As p increases, this

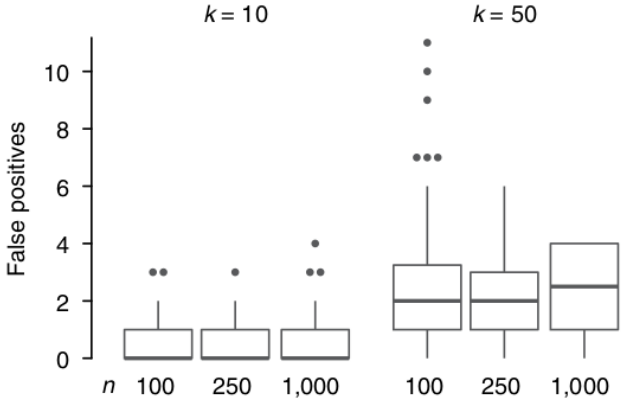


Fig. 3 | The number of false positives increases with each additional predictor. The box plots show the number of false positive regression-fit P values (tested at $\alpha = 0.05$) of 100 simulated multiple regression fits on various numbers of samples ($n = 100$, 250 and 1,000) in the presence of one true predictor and $k = 10$ and 50 extraneous uncorrelated predictors. Box plots show means (black center lines), 25th and 75th percentiles (box edges), and minimum and maximum values (whiskers). Outliers (dots) are jittered.

Correcting for multiple testing does not solve the problem of too many false-positive hits

Altman N, Krzywinski M. The curse(s) of dimensionality. Nat Methods. 2018 Jun;15(6):399-400. doi: 10.1038/s41592-018-0019-x. PMID: 29855577.

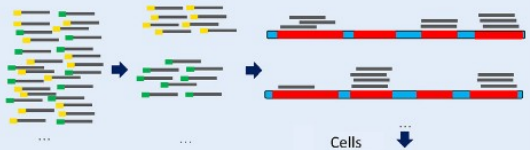
Main Steps in scRNAseq Analysis:

1) Quality Control
2) Removing technical variation
3) Dimension reduction
4) Clustering
5) Differential gene expression
6) Multi-Omics data integration
7) Etc. (trajectory, velocity, spatial...)

https://nbisweden.github.io/workshop-scRNAseq/

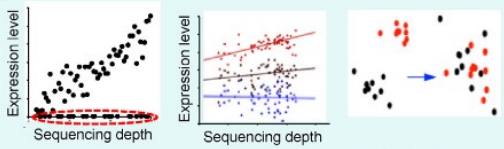
A typical scRNAseq workflow

1 Preprocessing



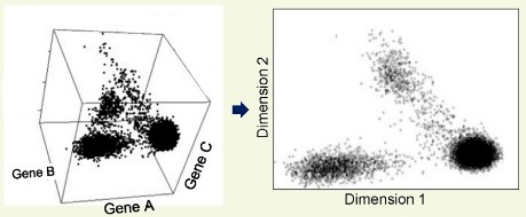
- Quality control
- Demultiplexing
- Deduplicating
- Trimming
- Mapping
- Quantification

2 Normalization



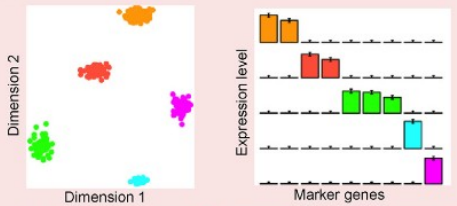
- Quantile normalization and scaling
- Imputation: MAGIC, SAVER, and scImpute
- Denoising data with internal control such as spike-ins and UMI: Seurat and Scater
- Correction of batch effects: ComBat, MNN, *etc.*

3 Dimensionality reduction



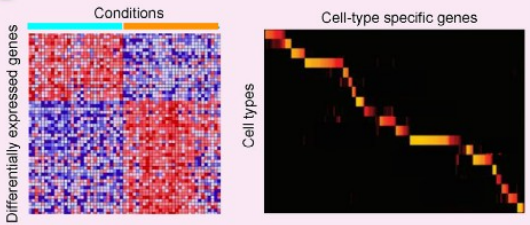
- Linear methods: PCA, ZIFA, ZINB-WaVE, *etc.*
- Non-linear methods: t-SNE, UMAP, SIMILR, *etc.*
- Deep neural network: scVI and DCA

4 Clustering



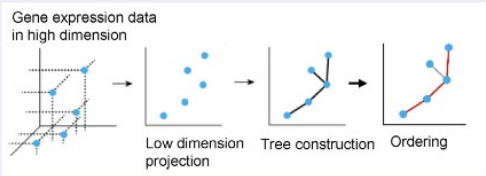
- Unsupervised: SNN-Cliq, RaceID, RaceID2, PhenoGraph, Seurat, and SC3
- Semi-supervised: Monocle and UNCURL
- Biclustering: BackSPIN

5 Differential gene expression



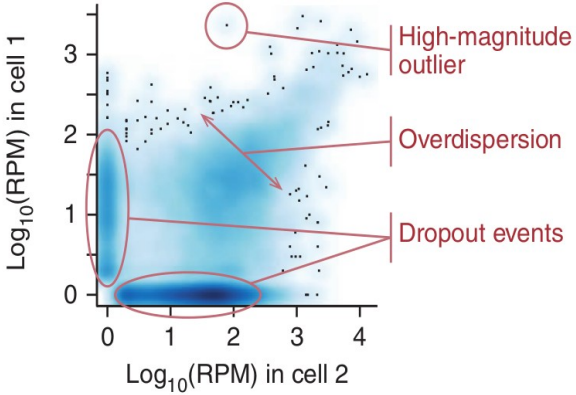
- Methods adopted from bulk RNA-seq: DESeq
- Prefiltering improves accuracy and robustness
- Mixture-model-based methods: MAST, scDE, and scDD

6 Gene expression dynamics



- Methods based on MST: Monocle and Waterfall
- Methods based on topological analysis: scTDA, Wishbone, SLICER, Destiny, and Sincell

Some steps are similar to bulk RNAseq, but the majority are not



Mu et al., Gen. Prot. Bio. 2019

Most Popular scRNAseq Workflows

SEURAT: Rahul Satija Lab



SCANPY: Fabian Theis Lab

scanpy

stable

Search docs

- Tutorials
- Usage Principles
- Installation
- API
- External API
- Ecosystem
- Release notes
- Community
- News
- Contributing
- Contributors
- References

Read the Docs v: stable

» Scanpy – Single-Cell Analysis in Python

Edit on GitHub

stars 1.3k | pypi v1.9.1 | downloads 1M | downloads 29k | docs passing | Azure Pipelines succeeded | discourse 1.8k posts | zulip join chat

Scanpy – Single-Cell Analysis in Python

Scanpy is a scalable toolkit for analyzing single-cell gene expression data built jointly with anndata. It includes preprocessing, visualization, clustering, trajectory inference and differential expression testing. The Python-based implementation efficiently deals with datasets of more than one million cells.

- Discuss usage on Discourse and development on GitHub.
- Get started by browsing tutorials, usage principles or the main API.
- Follow changes in the release notes.
- Find tools that harmonize well with anndata & Scanpy via the external API and the ecosystem page.
- Check out our contributing guide for development practices.
- Consider citing Genome Biology (2018) along with original references.

Key Contributors

anndata graph | scanpy graph | ⚛ = maintainer

- Isaac Virshup: lead developer since 2019 ⚛
- Gökcen Eraslan: developer, diverse contributions ⚛
- Sergei Rybakov: developer, diverse contributions ⚛
- Fidel Ramirez: developer, plotting ⚛
- Giovanni Palla: developer, spatial data
- Malte Luecken: developer, community & forum
- Lukas Heumos: developer, diverse contributions
- Philipp Angerer: developer, software quality, initial anndata conception ⚛
- Alex Wolf: lead developer 2016-2019, initial anndata & scanpy conception
- Fabian Theis & lab: enabling guidance, support and environment

News

Scanpy hits 100 contributors! 2022-03-31

100 people have contributed to Scanpy's source code!

Of course, contributions to the project are not limited to direct modification of the source code. Many others have improved the project by building on top of it, participating in development discussions, helping others with usage, or by showing off what it's helped them accomplish.

Thanks to all our contributors for making this project possible!

New community channels 2022-03-31

We've moved our forums and have a new publicly available chat!

- Our discourse forum has migrated to a joint scverse forum (discourse.scverse.org).
- Our private developer Slack has been replaced by a public Zulip chat (scverse.zulipchat.com).

SCRAN: John Marioni Lab

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scran

platforms all | rank 83 / 2140 | support 4 / 4 | in Bioc 6.5 years
build warnings | updated < 3 months | dependencies 59

DOI: 10.18129/B9.bioc.scran

Methods for Single-Cell RNA-Seq Data Analysis

Bioconductor version: Release (3.15)

Implements miscellaneous functions for interpretation of single-cell RNA-seq data. Methods are provided for assignment of cell cycle phase, detection of highly variable and significantly correlated genes, identification of marker genes, and other common tasks in routine single-cell analysis workflows.

Author: Aaron Lun [aut, cre], Karsten Bach [aut], Jong Kyoung Kim [ctb], Antonio Scialdone [ctb]

Maintainer: Aaron Lun <infinite.monkeys.with.keyboards at gmail.com>

Citation (from within R, enter `citation("scran")`):

Lun ATL, McCarthy DJ, Marioni JC (2016). "A step-by-step workflow for low-level analysis of single-cell RNA-seq data with Bioconductor." *F1000Res.*, **5**, 2122. doi: 10.12688/f1000research.9501.2.

Installation

To install this package, start R (version "4.2") and enter:

```
if (!require("BiocManager", quietly = TRUE))
    install.packages("BiocManager")

BiocManager::install("scran")
```

For older versions of R, please refer to the appropriate Bioconductor release.

Documentation

To view documentation for the version of this package installed in your system, start R and enter:

```
browseVignettes("scran")
```

HTML	R Script	Using scran to analyze scRNA-seq data
PDF		Reference Manual
Text		NEWS

Documentation »

Bioconductor

- Package vignettes and manuals.
- Workflows for learning and use.
- Several online books for comprehensive coverage of a particular research field, biological question, or technology.
- Course and conference material.
- Videos.
- Community resources and tutorials.

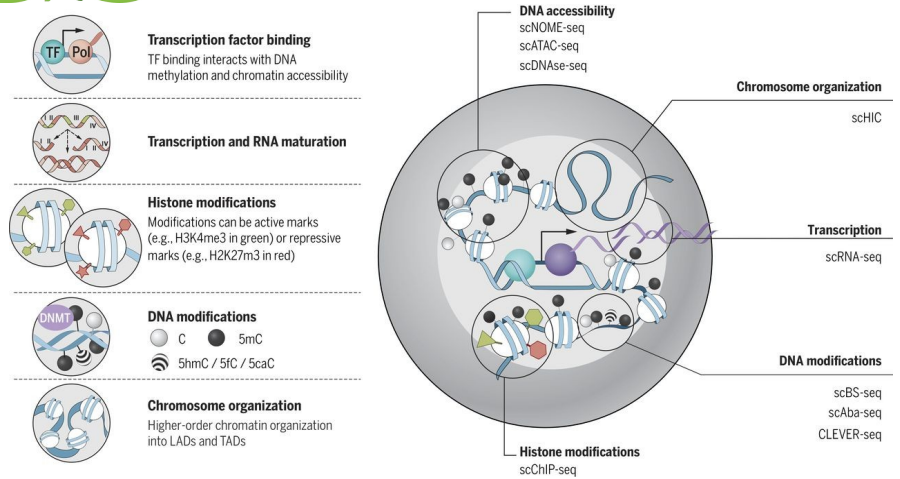
R / CRAN packages and documentation

Support »

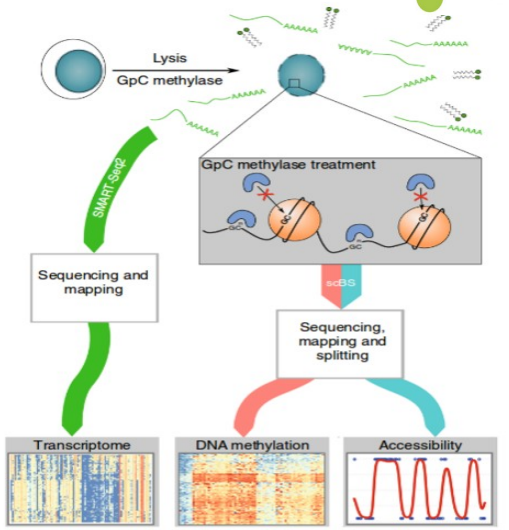
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- Support site - for questions about Bioconductor packages
- Bioc-devel mailing list - for package developers

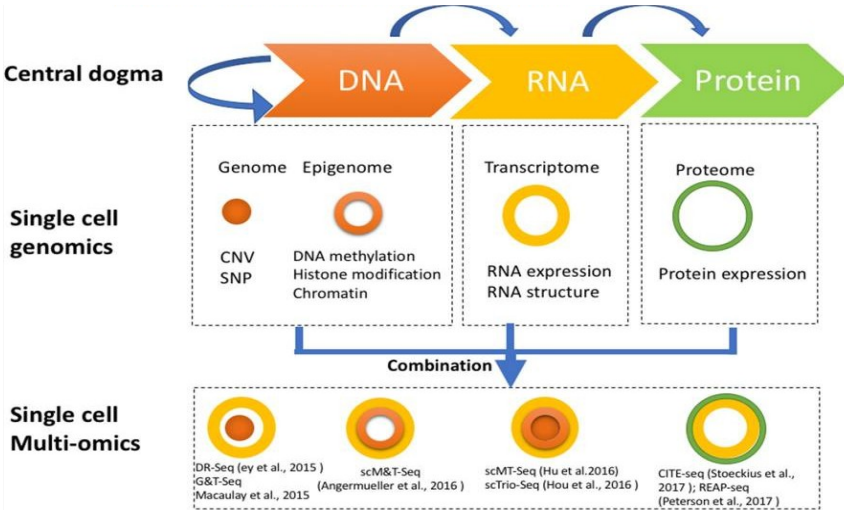
Multi-modal scOmics technologies



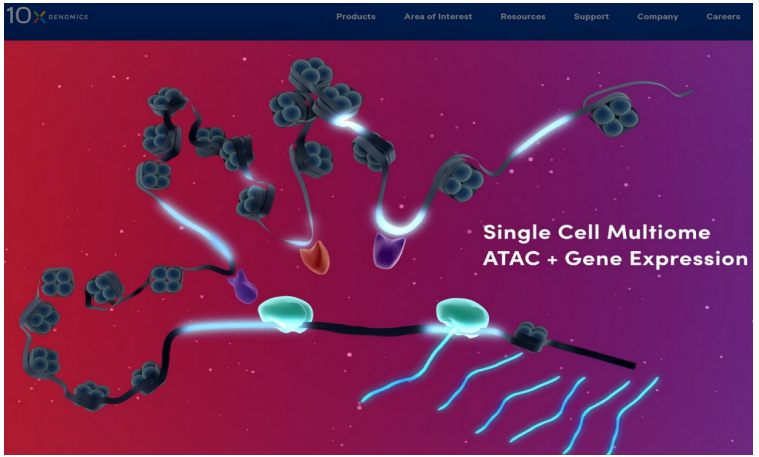
Kelsey et al., 2017, Science 358, 69-75



Clark et al., 2018, Nature Communications 9, 781



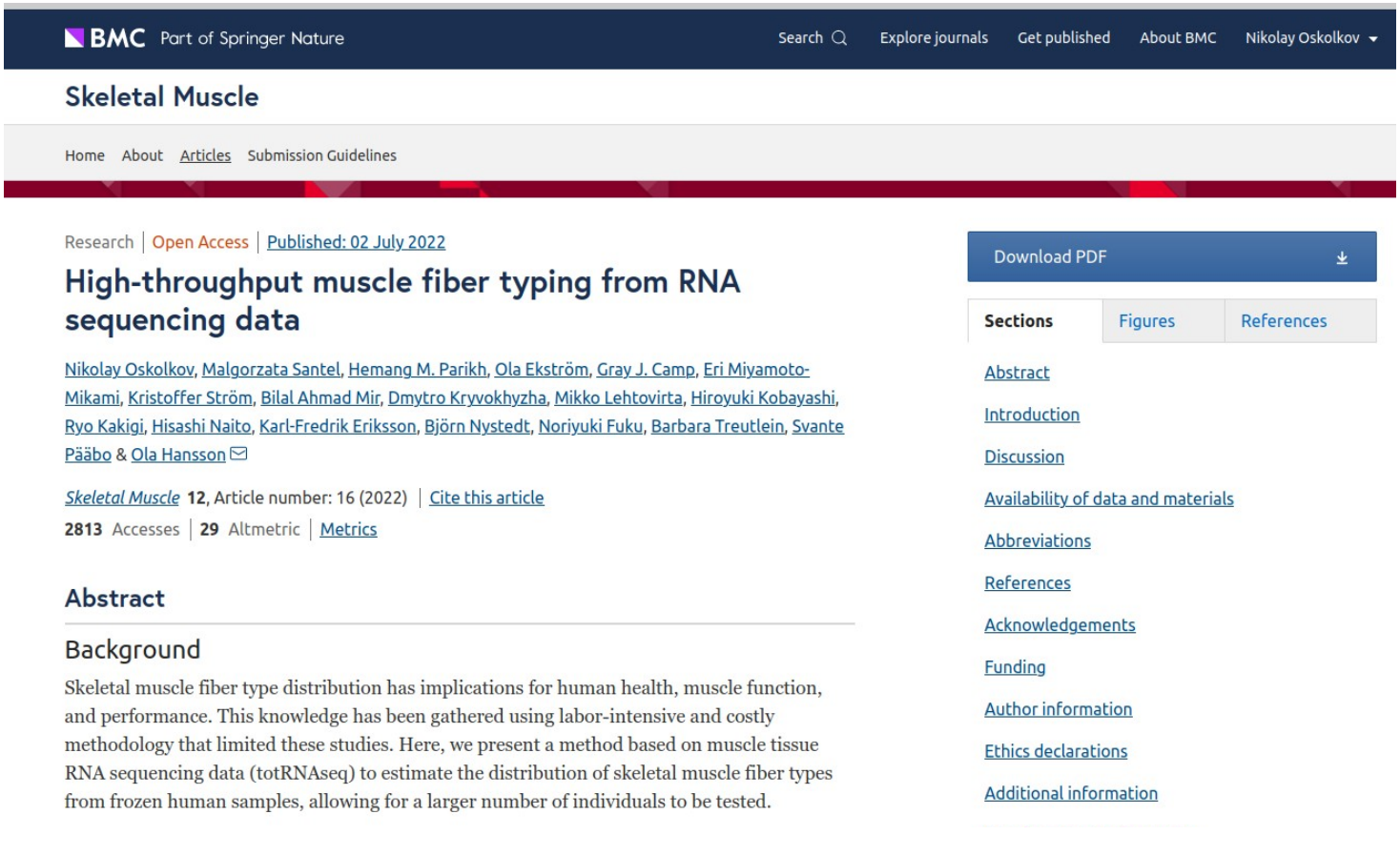
Hu et al., 2018, Frontier in Cell and Developmental Biology 6, 1-13



10X Genomics Multiome ATAC + Gene Expression

Practical Demonstration

https://github.com/NikolayOskolkov/SBW2022



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Skeletal Muscle

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High-throughput muscle fiber typing from RNA sequencing data

Nikolay Oskolkov, Malgorzata Santel, Hemang M. Parikh, Ola Ekström, Gray J. Camp, Eri Miyamoto-Mikami, Kristoffer Ström, Bilal Ahmad Mir, Dmytro Kryvokhyzha, Mikko Lehtovirta, Hiroyuki Kobayashi, Ryo Kakigi, Hisashi Naito, Karl-Fredrik Eriksson, Björn Nystedt, Noriyuki Fuku, Barbara Treutlein, Svante Pääbo & Ola Hansson

Skeletal Muscle **12**, Article number: 16 (2022) | Cite this article

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Abstract

Introduction

Discussion

Availability of data and materials

Abbreviations

References

Acknowledgements

Funding

Author information

Ethics declarations

Additional information

Abstract

Background

Skeletal muscle fiber type distribution has implications for human health, muscle function, and performance. This knowledge has been gathered using labor-intensive and costly methodology that limited these studies. Here, we present a method based on muscle tissue RNA sequencing data (totRNAseq) to estimate the distribution of skeletal muscle fiber types from frozen human samples, allowing for a larger number of individuals to be tested.

Session summary

Take home messages of the session:

1) Single cell technologies provide unprecendented Big Data, which is promising for machine learning analysis

2) The curse of dimensionality is not a problem for single cell analysis because of large sample sizes

3) Dropout effect is a peculiarity of single cell analysis

National Bioinformatics Infrastructure Sweden (NBIS)



Knut och Alice Wallenbergs Stiftelse

Vetenskapsrådet

Lunds Universitet

Acknowledgments: NIRI + TARGETWISE



European Regional Development Fund project “Functional Omics Analysis of Metabolic Diseases to Advance Drug Discovery Research Excellence in LIOS - TARGETWISE” (No.1.1.1.5./2/24/A/003)

2027 National Development Plan

Funded by the European Union

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